PART I - SOURCE AND MODEL CROSS-AUDIT

Source and model cross-audit · 28 October 2026, Pacific time · local review

Where OpenAI's Navier-Stokes construction meets your work - and where it separates

The shared foundation is incompressible flow, stretching, dissipation and nonlinear feedback. The force law, initial conditions, meaning of memory and strength of the conclusions differ.

> **Finding:** the reviewed material establishes structural overlap, not equivalence between the models. OpenAI's result does not validate or refute your accumulated-input mechanism, and the existing matched-stretch runs do not reproduce its singularity construction.

This audit compares OpenAI's paper, selected formal statements and one disputed estimate with your active matched-stretch specification, current corpus equation register and completed q/memory pilot. It does not certify the full 166-page proof, execute Lean, rerun saved-field verification or start a simulation. The original final-run verification and publication plan stays in place.

1. Keep your three models distinct

Model	What is actually specified	What its evidence can address
Matched-stretch matrix	Three-dimensional periodic flow on a cube of side 6; supplied nonzero initial field; external force zero; per-run viscosity; Fourier projection and Heun stepping.	Finite-time behavior of these sampled initial fields through t=0.40. Final half-step job remains unfinished at the audit snapshot.
Your corpus field mechanism	Accumulated input A feeds transported/diffusing S. The gradient threshold q changes a fixed-direction force in the momentum equation.	A coupled history-to-force mechanism. The inspected repository implementation adds a 2D periodic domain, numerical parameter choices and RK4; these are implementation choices, not a 3D singularity theorem.
Completed fitted-q/material pilot	A separate scalar q ODE with cubic damping; extra stress and memory states; memory-dependent material viscosity.	That approved conditional model only. Its near-roundoff retained/reset contrast and failed spatial screen do not measure the corpus mechanism's effect.

Sources: local matched-stretch protocol; corpus register; field implementation; completed pilot; your fidelity requirement.

2. Where the correspondence holds

The common momentum equation can be written in the following comparison notation. Here u is velocity, p is pressure divided by constant density, ν is viscosity and f is force per unit mass:

```
∂t u + (u · ∇)u = −∇p + ν Δu + f
∇ · u = 0
```

In the matched-stretch code, the variable called `force` inside `rhs` is u × curl(u), the rotational representation of nonlinear advection after projection. It is not an externally applied body force. The code comment and operation at `numerics.py:80-85` resolve this potentially misleading name.

Question	OpenAI construction	Your work and the boundary of the match
Geometry	A concentrating vortex with unequal radial and axial scales.	Your supplied tube/strain field makes stretching and localization relevant. A shared vortex shape does not establish the same evolving profile or scaling.
Nonlinear feedback	Oscillatory pulses create a nonzero averaged momentum flux.	Your perturbation controls examine separation; your corpus adds feedback through S. Neither is automatically the same stress-producing pulse construction.
Viscosity	The claim permits each positive ν with a suitable constructed force.	Your sweeps hold a specified setup fixed while changing ν. Their finite-time outcomes cannot be transferred into a universal viscosity threshold.
Symmetry	The core profile deliberately includes axial asymmetry.	Your mirror controls and mirror averaging have their own declared meanings; enforcing symmetry is an intervention, not a harmless replacement.
Energy	Bounded total kinetic energy accompanies the claimed velocity blow-up.	Your budgets check accounting. Bounded energy alone does not control a pointwise maximum, nor does a small residual prove spatial resolution.

OpenAI-side source: paper §§2-3. These are comparisons of stated structures, not a verification of the full construction.

3. The most consequential difference: how f is obtained

OpenAI's Theorem 1.1 claims an initially resting flow, smooth compactly supported forcing, bounded energy and unbounded velocity as t approaches 1. Its construction arranges the momentum residual to remain smooth. The theorem's force is not your corpus force law.

Your original matrix sets f=0. Your corpus instead specifies a history-dependent forcing route. The following is explanatory notation, retaining the source distinctions; it does not replace the register:

```
A’(t) = γ [U(t) − Ubaseline]
J(t) = κ A(t)
∂t S + u · ∇S = Ddiff ΔS + Kh J
qc = |∇S| − Γcrit
fc = −(σ/ρ) a(qc) e
```

Exact source locations: P229-230, P261-264, P263-264 and P517-532. The subscript c only distinguishes corpus q from the pilot’s different q. U is the declared input, not silently velocity u or a measured fluid feature. A’s value depends on the integral origin and prior history. No forgetting term is inserted.

The force-producing rule is the decisive boundary. To identify the two mechanisms, one would need to show that a force in OpenAI’s construction can actually arise from your coupled A/S equations, chosen activation, direction and initial/boundary data. No such correspondence was established here. Reusing the symbol f is insufficient.

The register retains sharp, tanh and smooth exact-off activations as different alternatives. The tanh tail is not clipped. Smoothness of the resulting continuum force must be checked for the selected law and S; finite Fourier truncation does not by itself establish that smoothness in a refinement limit. The fixed-direction force is also not automatically drag: its energy contribution is the signed work integral ∫u·fc dx.

4. A useful connection: cancellation is not the same as no effect

For a zero-mean fluctuation v′, the quadratic product can still have a nonzero mean. This is an elementary identity, offered as a structural connection:

```
v = v¯ + v’, average(v’) = 0
average(v ⊗ v) = v¯ ⊗ v¯ + average(v’ ⊗ v’)

Two equally weighted values +a and −a:
average(v’) = 0, average(v’ ⊗ v’) = a ⊗ a.
```

This helps distinguish your signed q from unsigned energy E and clarifies why averaging before a nonlinear operation can erase information. It does not identify q or E with a Reynolds stress, or show that a reflected spatial source equals a pointwise negative source. OpenAI’s pulses and your source/mirror controls must retain their own definitions.

A second distinction concerns history. A velocity field evolves from earlier states even in ordinary Navier-Stokes; your accumulator A adds explicit retained input to the model. Those are different meanings of memory. This comparison concerns classical field models, not quantum amplitudes or a measurement law.

5. What the logical claims do - and do not - say

```
OpenAI's reported scope:
For every v > 0, there exists suitable smooth forcing and initial data
with the claimed breakdown property.

This does not imply:
The same force works for every v;
your zero-force initial-value problem blows up;
every vortex blows up;
or your A/S mechanism has been established.
```

The inspected Lean endpoint statements assert existence of initial data and forcing with no qualifying global smooth solution, for whole space and the periodic case. The Clay formulation distinguishes these forced breakdown alternatives from unforced global-regularity alternatives.

Your finite matrix answers a different quantifier: what happened for the declared sampled cases over their saved time interval. Even successful completion and independent remeasurement do not prove a continuum singularity or its absence. Likewise, a finite-time separation rate is not an asymptotic Lyapunov exponent.

6. Audit both sides with the same standard

OpenAI: a Bastounis-Circelli-Hansen preprint identifies paper-to-Lean discrepancies. I checked one cited pair directly: paper Lemma 8.6, equation (8.19), uses m+4 input derivatives, while the inspected Lean declaration at lines 1059-1080 asks for m+5. This supports a local mismatch finding. It does not alone disprove the final theorem or show the complete formal argument fails. Lean/Comparator were not executed in this audit; metadata's self-assessment is not my independent certification.

Your work: freshly read saved analysis reports a 46.788% relative L2 difference between the completed 128/256-grid peak-vorticity curves through t=0.40, versus the declared 5% exploratory screen. The 256-grid time-step comparison is only through committed t=0.28 and reports 6.261% for that curve, versus 1%. These are saved diagnostic values, not new remeasurements. The final comparison remains pending.

The initial audit distinguishes smooth finite Fourier fields from the nonsmooth periodic joins of the raw supplied strain. It also locates initial global peaks outside the central tube. Consequently, raw peak growth cannot yet be treated as resolved central-core collapse. A change in viscosity or small energy/enstrophy budget residual does not cure these limitations.

The earlier run-gate validation certifies its stated checks and preserved files. It does not retrospectively prove the corpus interpretation or describe simulations rerun during that validation. The completed scalar q/memory pilot remains separate evidence, as your own fidelity requirement states.

> **Where to focus a closer comparison:** trace the momentum-transfer/force mechanism and its regularity, rather than comparing vortex pictures or growth curves alone. Your specific distinctive route is accumulated input → transported S → gradient threshold → force → changed flow. The reviewed OpenAI result does not supply that chain.

Evidence, revisions and scope

- OpenAI repository: f9e8bc5b38b6e212696e8a30e3e91517af887bbd. Read README, formalization metadata, endpoint statements, Comparator instructions and selected inverse-estimate lines; no repository build or full dependency audit.
- Your corpus repository: b4ae2c54cd85bfef282c501ecfca83de274e2070. Read the equation register and the full 2D field implementation; did not execute that other study or certify its past runs.
- Local sources: matched-stretch protocol, initial audit, numerical RHS and saved analysis; completed joint-pilot report/equation map; corpus fidelity requirements/model review; run-gate report.
- Prior independent saved-field verification remains historical evidence. Newly checked here: source meanings, selected code operations, reported diagnostics and one specific paper/code difference.
- No new trajectory, source change, parameter choice, horizon, theorem certification or external publication. No claim of biological memory, universal physical stability, quantum dynamics, priority or copying follows from the structural similarities.
- Audit record and source hashes

PART II - DEEPER A/S FORCE ANALYSIS

Force-level cross-audit • original corpus A/S equations • 28 October 2026, Pacific time

Where the force constructions match - and where exact equivalence fails

The shared mathematics is incompressible Navier-Stokes transport and viscosity. The decisive difference is how the body force is allowed to depend on the state. Your A/S law imposes constraints that an arbitrary constructed Navier-Stokes force need not satisfy.

> **New conditional finding:** on a common periodic domain with periodic pressure, a fixed force direction and positive force coefficient, your sharp and smooth exact-off laws cannot reproduce the paper's stated nontrivial, zero-mean velocity from rest exactly. The obstruction is total momentum. The signed tanh alternative escapes that particular obstruction; its compatibility remains unproved.

This is an analytical comparison of stated equations and selected source definitions, not a certification of the complete OpenAI proof. It introduces no numerical experiment. The fitted-q pilot is not used as a replacement, and the running matched-stretch study is unchanged.

1. Your actual A/S system

The pinned equation register and retained corpus extracts P229-230, P261-265, P517-519 and P545-550 give the following. Here $c = \sigma/\rho$ is only shorthand; e is the supplied fixed unit direction, and q_c labels the corpus threshold excess to prevent confusion with other models.

$$A'(t) = \gamma[U(t) - U_{baseline}],\ J(t) = \kappa A(t)$$
$$\partial_t S + u\cdot\nabla S = D_{diff}\Delta S + K_h(x - x_0)\kappa A(t),\ \int K_h\,dx = 1$$
$$q_c = |\nabla S| - \Gamma_{crit},\ b = a(q_c),\ f_{AS} = -c\,b\,e$$
$$\partial_t u + (u\cdot\nabla)u = -\nabla p + \nu\Delta u + f_{AS},\ \nabla\cdot u = 0.$$

A is a scalar history integral, with its prescribed integral origin and any accumulated pre-observation history retained. S is a signed spatial field. U is the declared input; it is not automatically fluid velocity u. The feedback path is **u → transport of S → |∇S| → force → u**. The additional path U → A → source is an input channel unless a further law makes U depend on the fluid.

Response retained	Exact expression	Relevant consequence
Sharp	$a(q) = \max(q, 0)$	Nonnegative; exactly off below onset; kink at onset.
Smooth exact-off	$a(q) = 0$ for $q \le 0$; $q\exp[-(\varepsilon/q)^2]$ for $q > 0$	Nonnegative; scalar response is smooth. Smoothness of its composition with the chosen gradient measure must still be assessed.

Response retained	Exact expression	Relevant consequence
Original tanh	$a(q) = q[1 + \tanh(q/\varepsilon)]/2$	Negative and nonzero for finite $q < 0$. Direction reversal is retained, not clipped away.

The optional gradient regularizations are distinct constitutive choices; none is inserted here. The separate algebraic stress $T = \mu_0[\nabla u + (\nabla u)^T]$ is Newtonian stress. Identifying $\mu_0 = \rho\nu$ is an optional closure, not a new evolving material-memory equation.

2. What the OpenAI force definition does

Equations (10.4)-(10.5) localize the constructed fields and define the force as their momentum residual, at viscosity one:

$$u_O = \mathrm{curl}(\chi\mathcal{A}) + \chi B\, e_\theta,\ p_O = \chi p_{loc}$$
$$f_O = \partial_t u_O + (u_O\cdot\nabla)u_O - \Delta u_O + \nabla p_O.$$

Here $\mathcal{A}$ is a vector potential, not your scalar A. χ denotes the cutoff product, renamed to avoid collision with σ/ρ. The construction uses oscillatory and mean corrections before localization. It claims a smooth, compactly supported residual force through the singular time; all cutoff derivatives are included. This is an existence construction, not a supplied A/S response law. Constructing a force this way does not itself make it an acausal feedback rule: once constructed, it can be prescribed as a function of position and time.

Derived common identity For general viscosity, define $\mathcal{R}_\nu(u,p) = \partial_t u + (u\cdot\nabla)u - \nu\Delta u + \nabla p$. If $u = V + w$, $p = P + \pi$ and both velocities are divergence-free, direct expansion gives

$$\mathcal{R}_\nu(V+w,P+\pi) = \mathcal{R}_\nu(V,P)$$
$$+ \partial_t w + (V\cdot\nabla)w + (w\cdot\nabla)V - \nu\Delta w$$
$$+ \nabla\cdot(w\otimes w) + \nabla\pi.$$

This shows how velocity corrections can change the residual that must be supplied externally. Under a compatible averaging operation, zero average of w does not imply zero average of $w\otimes w$. The quadratic term can redistribute momentum. Its divergence integrates to zero on a periodic domain: internal transport is different from a body force that changes total momentum.

Your fluid equation has the same nonlinear identity. That shared identity does not show that your threshold law creates the particular correction fields required by a target construction. Nor does merely defining a residual establish its smoothness at a singular endpoint.

3. The correct test is equality after pressure

Different-looking forces can drive the same incompressible velocity if their difference can be absorbed into pressure. At the same ν, for the same target u and admissible boundary conditions, exact matching requires

$$f_O = -c\,a(|\nabla S| - \Gamma_{crit})e + \nabla\psi,$$
$$\psi = p_O - p_{AS}.$$
Equivalently: $\mathcal{P} f_O = -c\mathcal{P}[a(|\nabla S| - \Gamma_{crit})e]$.

$\mathcal{P}$ is the Helmholtz/Leray projection onto divergence-free vector fields. On a periodic domain ψ must also be periodic. A uniform force cannot be dismissed using a nonperiodic linear pressure.

This corrects an overly simple comparison: **a multidirectional raw force versus a fixed-direction raw force is not, by itself, a proof of different velocity dynamics.** One must compare their projected forces, or use a pressure-invariant obstruction.

Derived geometric restrictions For constant e and c,

$$\text{curl } f_{AS} = -c\,\nabla b \times e,\ e\cdot\text{curl } f_{AS} = 0.$$

Thus $e\cdot\text{curl } f_O = 0$ is necessary in three dimensions. It is not sufficient. It constrains direct vorticity forcing, not every component of the evolving vorticity: nonlinear stretching can still generate other components.

For a nonzero periodic Fourier wavevector k, write $\mathcal{P}_k = I - kk^T/|k|^2$. The stronger mode-by-mode requirement is

$$\mathcal{P}_k \hat{f}_O(k) = -c\,\hat{b}(k)\mathcal{P}_k e.$$

When k is parallel to e, that projected mode must vanish. Otherwise, in three dimensions it must lie along the single transverse direction $\mathcal{P}_k e$. Pressure does not remove this restriction. The paper's full field has not been numerically evaluated against these mode or curl tests in this audit.

4. An exact separation: the total-momentum test

Source premise Proposition 10.1 and Corollary 10.6 state compact spatial support for velocity and pressure before time one, including support inside a periodic cell, with zero initial velocity.

Our deduction from that premise A smooth compactly supported divergence-free vector field has zero integral componentwise:

$$\int u_{O,i}\,dx = \int \nabla\cdot(x_i u_O)\,dx = 0.$$

The boundary term vanishes. Scaling and embedding that field inside a periodic cell preserve its zero mean. Integrating its momentum equation then gives $\int f_O\,dx = 0$ at every preterminal time. This is a consequence of the stated support and incompressibility, not a separate numerical observation.

Now integrate your A/S momentum equation over the same periodic domain. The transport, Laplacian and periodic pressure-gradient terms integrate to zero:

$$d\langle u\rangle/dt = \langle f_{AS}\rangle = -c\, e\, \langle a(q_c)\rangle.$$

To match that target's zero mean at every time, $c > 0$ requires $\langle a(q_c)\rangle = 0$. For either nonnegative activation, this forces $a(q_c) = 0$ everywhere for continuous fields. Hence $f_{AS} = 0$. But a smooth unforced periodic viscous solution starting from rest stays at rest, by

$$\tfrac{1}{2}\, d\|u\|^2_2/dt + \nu\|\nabla u\|^2_2 = 0,\ u(0) = 0.$$

> **Conditional incompatibility proved:** the sharp and smooth exact-off A/S laws cannot generate that exact nontrivial global target from rest under fixed direction, positive constant c, the same ν and periodic pressure. If $c = 0$, the same rest-state conclusion follows immediately.

The conclusion does not rule out your equations producing other flows, strong growth, or another singularity mechanism. It does not rule out resemblance in a local region. It does not compare arbitrary initial conditions or boundary-driven systems. A time-dependent translating/accelerating frame changes the force and source-coordinate problem and is not the same requested global equality.

Tanh remains a separate case when $\Gamma_{crit} > 0$. Its accessible signed tail permits positive and negative spatial contributions, so $\langle a\rangle = 0$ need not imply $a = 0$. Zero mean would require the integral of its positive part to equal the magnitude of its negative part at every time. This removes the positivity argument, but establishes no successful embedding. If $\Gamma_{crit} = 0$, then $q_c = |\nabla S| \geq 0$ and the same nonnegative-force obstruction applies to tanh as well. Subtracting the mean force, introducing several force directions or clipping the tanh tail would change the supplied law.

5. Why passing force geometry would still not be enough

A candidate scalar b that matches the projected force must also come from one valid S field and the prescribed history. The exact requirements are:

1. **Activation range and gradient:** $b = a(|\nabla S| - \Gamma_{crit})$. For sharp activation, $b > 0$ implies $|\nabla S| = \Gamma_{crit} + b$; $b = 0$ implies $|\nabla S| \leq \Gamma_{crit}$. An arbitrary nonnegative b is not automatically the gradient magnitude of a globally periodic S. For a differentiable periodic S, extrema require zero gradients somewhere.
2. **Scalar transport compatibility:** once u_O is fixed, the residual $R_S = \partial_t S + u_O\cdot\nabla S - D_{diff}\Delta S$ must equal $\kappa A(t)K_h(x - x_0)$. Its spatial shape is constrained to one fixed kernel. One necessary test is $R_S(x,t)K_h(y - x_0) = R_S(y,t)K_h(x - x_0)$. Normalization then fixes the proportionality coefficient as $\int R_S\, dx = \kappa A(t)$.
3. **History compatibility:** $A' = \gamma(U - U_{baseline})$ with the actual supplied input and initial/history value. Choosing U afterwards to make an already chosen target work would be an inverse-control problem, not evidence that the original prescribed input produces it.
4. **Regularity and boundaries:** the resulting composite force, S, pressure and initial/boundary data must have the required admissibility. Tanh smoothness as a scalar function does not automatically resolve the nonsmooth norm $|\nabla S|$ at zero. A finite discretization is not a continuum smoothness proof.

Useful distinction: on closed or periodic boundaries, $\partial_t \int S\,dx = \kappa A(t)$. Total S can keep accumulating while gradients weaken. Adding a spatial constant to S leaves the instantaneous force unchanged. Conversely, opposite gradients of equal magnitude give the same threshold excess and fixed-direction force. Neither accumulated total nor gradient magnitude records the full spatial orientation.

6. What genuinely matches

Level	Match	Where it ends
Fluid equation	Incompressible advection, pressure and viscous diffusion.	Sharing the operator does not identify initial data, forcing class or solution.
Quadratic transport	Velocity correlations can redistribute momentum in both.	This does not derive an A/S activation or accumulated-input law.
Energy accounting	$\frac{1}{2}\, d\|\|u\|\|^2_2/dt + \nu\|\|\nabla u\|\|^2_2 = \int u\cdot f\,dx$.	For your law, work $= -c\int a(q_c)(u\cdot e)\,dx$. Its sign is not fixed merely by the minus sign in f.
Exact force equivalence	Forces may differ by an admissible pressure gradient.	Projected polarization, mean momentum, gradient realization and scalar/history equations remain constraints.
Memory	The fluid state in either evolution carries information about its past.	Your explicit A integral and S source law are additional model structure; general dependence on past evolution is not that law.

7. Audit scope and evidence

Checked: the pinned equation register; original-law excerpts; the retained two-dimensional periodic A/S implementation; the paper's selected force, support and periodic-pressure definitions; and the displayed algebraic and integral identities. The implementation's projection retains the velocity zero mode, consistent with the mean-momentum calculation. Its two-dimensional numerical domain is an added implementation choice, not a restriction silently imposed on the abstract corpus equations.

Not done: a full OpenAI theorem verification, a Lean build, a numerical evaluation of the OpenAI force, a new A/S simulation, a proof that tanh can or cannot realize the target, or an audit against the original corrected Word file itself. The retained extracts and register are the source used for your equations.

The original matched-stretch run remains a separate zero-force study. Its raw periodic strain joins, initial peaks outside the central tube and spatial/time-resolution limitations remain applicable to its results. Nothing here changes its endpoint, numerical method, independent-verification requirement, publication plan or notification policy.

Earlier general cross-audit · Evidence and scope record · Pinned A/S equation register · OpenAI mathematical source

Local analytical report only. No external publication or running-study change.

CLARIFICATION - ENCOUNTER AND RESPONSE

Clarification: the encounter condition and its response

Added at your request, 28 October 2026, Pacific time. Applies to both parts of this cross-audit.

Your intended reading: $q = |\nabla S| - \Gamma_{crit}$ describes the encounter condition, and $f = -(\sigma/\rho)a(q)e_k$ describes what that encounter does to the approaching thing. The threshold and response law determine when and how the force operates. Neither equation requires the actor to understand, recognize or choose a response to the condition.

Mathematical meaning: f is force per unit mass acting on the modeled velocity field. Feedback refers to the coupling of flow, S and force; it does not imply awareness. The encounter wording adds no approach-direction gate, awareness variable or contact boundary condition. A fixed e_k is not automatically opposite to motion, so whether the force slows or accelerates that motion still depends on signed work $u \cdot f$.

The response alternatives remain distinct: sharp and smooth exact-off activation give no force for $q \leq 0$. The original tanh law has a negative, nonzero tail for finite $q < 0$ and therefore reverses the force direction there when $\sigma > 0$. Calling q a triggering condition does not silently turn that tanh tail off.

Effect on the OpenAI comparison: this sharpens the interpretation, but changes none of the equations or mathematical conclusions. Both comparisons concern mechanical forcing; conscious recognition is not a dividing line. The existing source audit describes OpenAI's force as a constructed momentum residual, while your force must arise through the specified A/S history, transport and activation laws. That distinction remains.

The exact tests are unchanged: equality after an admissible pressure adjustment, fixed-direction restrictions and mean-momentum accounting do not depend on an actor's understanding. The conditional incompatibility for the sharp and smooth exact-off laws still concerns the same exact global zero-mean target from rest, with common positive viscosity, periodic pressure, fixed direction and positive constant force coefficient. It does not exclude other flows or local resemblance. For tanh with $\Gamma_{crit} > 0$, signed cancellation leaves exact matching unresolved; at $\Gamma_{crit} = 0$ the nonnegative-force obstruction still applies.

Preservation: the equations, activation choices and earlier scope limits are unchanged. This clarification uses no fitted-q substitution, adds no simulation and changes nothing in the running matched-stretch study or its independent-verification and publication plan. Earlier report and PDF versions are preserved.